

Real-Time 3-D Imaging Using an Air-Coupled Ultrasonic Phased-Array

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Abstract—We present an air-coupled ultrasonic imaging system based on a 40-kHz 8×8 phased-array for 3-D real-time localization of multiple objects in the far-field. By attaching a waveguide to the array, the effective interelement spacing is reduced to half wavelength. This enables grating lobe-free transmit and receive beamforming with a uniform rectangular array of efficient low-cost transducers. The system further includes custom transceiver electronics, an field programmable gate array (FPGA) system-on-chip and a PC for GPU accelerated frequency domain signal processing, consisting of matched filtering, conventional beamforming, and envelope extraction using Nvidia Compute Unified Device Architecture (CUDA) and OpenGL for visualization. The uniform rectangular layout allows utilizing multiple transmit and receive methods, known from medical imaging applications. Thus, the system is dynamically adaptable to maximize the frame rate or detection range. One implemented method demonstrates the real-time capability by transmitting a hemispherical pulse (HP) with a **single transducer to irradiate the surroundings simultaneously**, whereas all transducers are used for echo reception. The imaging properties, such as axial and lateral resolution, field of view and range of view, are characterized in an anechoic chamber. The object localization is validated for a horizontal and vertical field of view of $\pm 80^\circ$ and a range of **view of 0.5–3 m with 29 frames/s**. Using the same system, a comparison between the HP method and the dynamic transmit beamforming method, which transmits multiple sequential beamformed pulses for long-range localization, is provided.

Index Terms—Air-coupled, imaging, phased-array, sonar, ultrasound.

I. INTRODUCTION

ANY automation task that requires environmental monitoring relies on 3-D imaging systems. These systems are commonly based on a combination of LiDAR [1], radar [2], [3] and camera sensors [4]. As a complement, ultrasonic sensors are utilized as 1-D range finders [5], [6] due to their robust operation in poor visibility, contamination, and in the presence of optically reflective or transparent surfaces. However, these 1-D sensors provide only limited directional information. The combination of multiple ultrasonic

transducers in a phased-array arrangement, as used in **medical sonography** [7] and underwater sonar [8], [9], allows the localization of objects in three dimensions, i.e., range, azimuth, and elevation angles. **We propose that introducing ultrasonic phased-array imaging to in-air applications improves the robustness for object localization in conjunction with the abovementioned sensors.**

However, compared to medical sonography, there are three major differences for air-coupled ultrasound. **First, sound propagates more than four times slower in air than in water or tissue [10].** Second, a much longer range of view is required for object detection. **Third, the signal frequency must be in the lower ultrasonic frequency range to avoid strong attenuation by air.** These differences imply that most of the region of interest is in the far-field and the achievable frame rate is considerably limited. In this context, we evaluate an air-coupled ultrasonic phased-array system by investigating its 3-D imaging properties, as well as its real-time capability.

The idea of combining ultrasonic transducers in a phased-array has been applied in a variety of other in-air applications, **including haptic hologram generation [11], [12], levitation of small objects [13], [14], highly directional parametric loudspeakers [15], [16], surface wave elastography [17], vortex generation [18], and power transfer [19], [20].** In these studies, efficient piezoelectric ultrasonic transducers (Murata MA40S4S) with a resonant frequency of 40 kHz have been chosen as array elements.

However, arranging these transducers in an array is problematic. The diameter of this transducer type is too large to meet the $\lambda/2$ criterion for the maximum interelement spacing, resulting in high intensity sound emission in unintended directions, also referred to as grating lobes [21]. Although grating lobes do not hinder the operation of the in-air applications mentioned, they may cause harm to nearby users [22]. In contrast, for acoustic imaging the avoidance or suppression of grating lobes is mandatory in order to locate objects without ambiguities.

Different techniques have been applied for unambiguous imaging with air-coupled ultrasonic phased-arrays. In [23] and [24], a movable transducer array mounted on a 2-D positioning system is used to form a synthetic aperture [25]. In [26], the array layouts for pulse transmission and echo reception differ, resulting in grating lobes being canceled out in the point spread function. Kumar and Furuhashi [27] have

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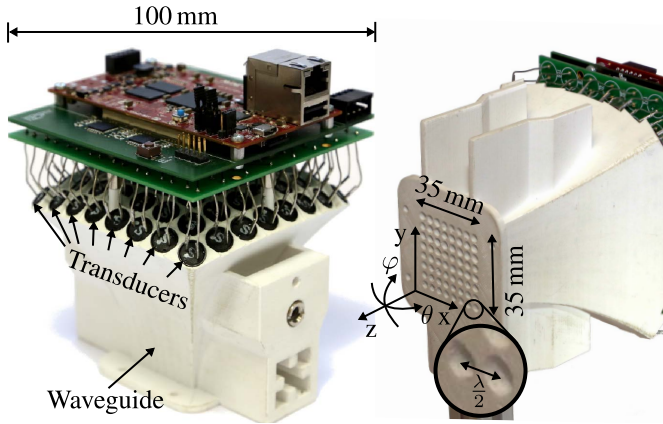


Fig. 1. Uniform rectangular 8×8 phased-array with 3-D-printed waveguide and electronics attached. The ultrasonic transducers are inserted into the waveguide reducing the effective interelement spacing to $\lambda/2$ for grating lobe-free beamforming.

followed a similar approach by using a transmitter array with Murata MA40S4S transducers and a receiver array consisting of small microelectromechanical system (MEMS) microphones. Steckel *et al.* [28] and Kerstens *et al.* [29] used a single broadband transducer to transmit a chirp pulse and a random MEMS microphone array for echo reception. Since only the direction of the main lobe of the received directional pattern is frequency independent, grating lobes are suppressed in the directional energy spectrum after matched filtering. A more conservative approach is to meet the $\lambda/2$ criterion. This has been accomplished by developing small size ultrasonic transducers based on PMUTS [30], [31], CMUTS [32]–[34], PVDF materials [35]–[37], and ferroelectrics [38]. However, lowering the operation frequency while maintaining efficiency regarding the transmitted sound pressure level has been challenging. Instead of using smaller transducers, Takahashi *et al.* [39] attached shrinking tubes functioning as waveguides to the efficient low-cost Murata MA40S4S transducers, reducing the effective interelement spacing.

Based on this idea, we have created a 3-D printed waveguide containing 64 acoustic channels into which MA40S4S transducers are inserted [40]. This structure forms a uniform rectangular array (URA) with $\lambda/2$ spacing (Fig. 1), enabling grating lobe-free transmit and receive beamforming, which has also been utilized for flow measurement [41] and non-destructive, non-contact testing (NDT) based on Lamb waves [42]. The URA layout allows the adaptation of multiple transmit and receive methods, commonly used in medical imaging, e.g., conventional dynamic transmit beamforming (DTB) for achieving high SNR and resolution. Other techniques are aimed at increasing the frame rate by widening the transmit beam, thus reducing the number of required pulses for 3-D image generation, e.g., fan-beam scanning [43] and diverging wave transmission [44], [45]. Therefore, by utilizing different methods, our air-coupled system is dynamically configurable to meet varying requirements.

In order to investigate the capabilities of the system, we have adapted two different imaging methods, each addressing one limitation of air-coupled ultrasound, i.e., the limited range of

view and frame rate. In our previous work [46], the DTB method has been validated for object localization up to a range of 6 m, limited by the size of the anechoic chamber. In this method, a pulse-echo detection is performed sequentially for all directions of the discretized region of interest using all transducers for beamformed pulse transmission and echo reception. Although this technique enables long range localization, requiring multiple pulse-echo detections results in slow image generation.

In this article, we evaluate the real-time imaging capability of the system by choosing a method that maximizes the frame rate. Since fan-beam scanning and the diverging wave technique still require multiple pulses per image, we use only one transducer to transmit a single hemispherical pulse (HP) to irradiate the surroundings simultaneously, at the expense of range and resolution. The specific characteristics of the system when using the DTB and HP method are then compared.

The remainder of this article is structured as follows. In Section II, the array system is introduced, including the waveguide and transducer arrangement, as well as the transceiver electronics and the PC. In Section III, the processing stages of the HP method, i.e., the discretization of the region of interest, the pulse-echo sequence, the array signal processing, and the image visualization are described. Here, insight into the implementation using Compute Unified Device Architecture (CUDA) and GPU acceleration and its advantages are provided, as well. In Section IV, the HP method is evaluated experimentally and compared to the DTB method concerning the imaging properties and achievable frame rates.

II. SYSTEM OVERVIEW

The phased-array system includes four major components, which are the phased-array itself, the custom transceiver electronics, an field programmable gate array (FPGA) board based on the Xilinx Zynq system-on-chip (SoC) and a conventional PC (Fig. 2).

The phased-array consists of $M = 64$ piezoelectric transducers (Murata MA40S4S) with a resonant frequency of $f_0 = 40$ kHz, which are arranged in an 8×8 URA. A waveguide with 64 equal-length channels is attached to the transducers, reducing the receiving aperture to $35 \text{ mm} \times 35 \text{ mm}$ and the effective interelement spacing to half wavelength $d = \lambda/2 = 4.3 \text{ mm}$ for grating lobe-free beamforming (Fig. 1). The position of the m th channel port is given by

$$[x_m, y_m, z_m] = [d(m \bmod 8), d\lfloor m/8 \rfloor, 0] \quad (1)$$

where mod denotes the modulo operator, $\lfloor \cdot \rfloor$ denotes the floor function, $m \in [0, \dots, M-1]$ is the channel index and d is the interelement spacing.

The transceiver electronics is capable of driving all ultrasonic transducers with individually phase-shifted transmit signals. For this, eight multichannel MOSFET pulser ICs (Microchip HV7355) are used, each receiving one digital serial signal and generating eight unipolar square-wave signals with the driving amplitude of $20 V_{pp}$. However, only one ultrasonic transducer driven in the HP method is described in this article. In order to toggle from transmit to receive mode,

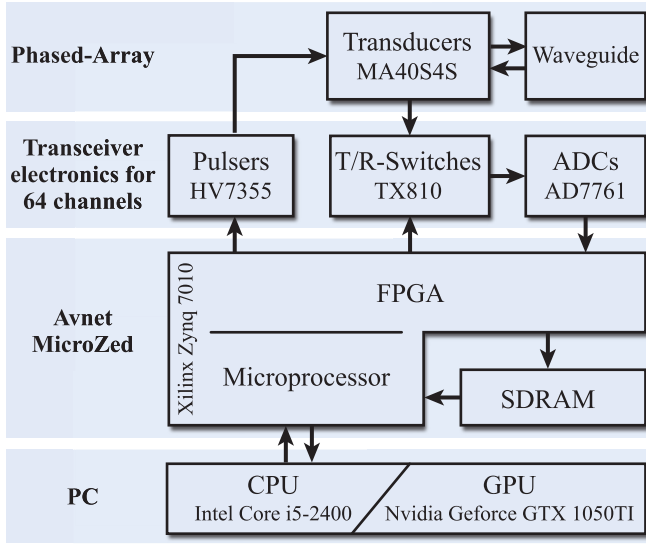


Fig. 2. Phased-array system components. The transceiver electronics includes eight pulser ICs each capable of driving up to eight transducers. However, only one transducer is driven in the HP method. Sixteen TR-switches toggle between transmit and receive mode. The FPGA of the Xilinx Zynq SoC generates the transmit signal, controls the pulse-echo sequence and receives the digitized data of the eight multichannel ADCs. The microprocessor receives control commands from the PC via gigabit Ethernet and, likewise, returns the digitized data. Array signal processing and visualization are performed by the GPU.

16 four-channel T/R switches (Texas Instruments TX810) are activated and the outputs of the pulser ICs are set to high impedance. In receive mode, the incoming signals of all transducers are digitized in parallel by eight multichannel 16-bit ADCs (Analog Devices AD7761) with a sampling rate of $f_s = 200$ kSa/s per channel. The ADCs can be synchronized and have a serial interface reducing FPGA pin usage for the data reception.

The FPGA board (Avnet MicroZed) features 1 GB of shared SDRAM and one Xilinx Zynq 7010 SoC containing an FPGA and a microprocessor in one package. The FPGA is used for generating the transmit signal, sequence control, as well as reception and storing of all digitized data of the ADCs. The microprocessor receives control commands from the PC via Ethernet and, likewise, returns the digitized raw data.

On the PC, a custom C++ program is executed for system control via user interface, communication, signal processing, and visualization. The latter two tasks are handled by a GPU (Nvidia Geforce GTX 1050 TI) using the Nvidia CUDA toolkit and OpenGL. The CUDA toolkit provides ready-to-use batched FFT functions, thus enabling the implementation of parallel low-latency signal filtering and efficient beamforming in the frequency domain. In general, the system architecture, which is similar to software-based medical open scanners [47], offers a number of advantages over hardware-based signal processing using FPGAs. First, GPUs are quickly reprogrammable without requiring time-consuming logic synthesis, thus accelerating development cycles. Second, since the processed data is available in the GPU memory, additional parallel image processing, e.g., object tracking, can be applied directly. Furthermore, the option of exporting raw data to frameworks,

such as MATLAB, simplifies prototyping of advanced imaging algorithms.

In summary, the system is easily reconfigurable and expandable, yet still maintaining high processing speed through parallelization. In Section III, the generation of an acoustic image is described.

III. THEORY OF OPERATION

In order to generate an acoustic image showing the positions of objects in front of the phased-array, three processing stages are performed. First, in the pulse-echo sequence, an ultrasonic pulse is transmitted and the induced echo signals are captured. Second, using array signal processing, the originating positions and amplitudes of the echoes are determined from the received signals. Third, the obtained position and amplitude information are visualized in a horizontal sectional image (B-Scan) and a 3-D volume image (3-D-Scan). These three steps are repeated to generate multiple sequential images, which are referred to as frames.

Before starting the frame generation, the region of interest is discretized and defined with the field of view $\mathcal{P}$ and the range of view $\mathcal{R}$. The field of view is a discrete set of directions $\mathbf{p}_k = (\theta_k, \varphi_k)$ to be considered for echo localization and visualization, that is,

$$\mathcal{P} = \{(\theta_0, \varphi_0), \dots, (\theta_{K-1}, \varphi_{K-1})\} \quad (2)$$

where K is the number of directions, θ is the azimuth angle and φ is the elevation angle. The discretization of the field of view is specified with the angular boundaries $\theta_{\min}, \theta_{\max}, \varphi_{\min}, \varphi_{\max}$ and the angular step sizes $\theta_{\text{step}}, \varphi_{\text{step}}$. The number of directions K is determined with

$$K_\theta = \frac{\theta_{\max} - \theta_{\min}}{\theta_{\text{step}}} + 1, \quad K_\varphi = \frac{\varphi_{\max} - \varphi_{\min}}{\varphi_{\text{step}}} + 1 \quad (3)$$

$$K = K_\theta \cdot K_\varphi \quad (4)$$

where K_θ and K_φ are the number of different azimuth and elevation angles, respectively. The angles corresponding to the k th direction $\mathbf{p}_k$ are given by

$$\mathbf{p}_k = \begin{bmatrix} \theta_k \\ \varphi_k \end{bmatrix}^T = \begin{bmatrix} \theta_{\min} + (k \bmod K_\theta) \cdot \theta_{\text{step}} \\ \varphi_{\min} + \lfloor k/K_\theta \rfloor \cdot \varphi_{\text{step}} \end{bmatrix}^T \quad (5)$$

where $k \in [0, \dots, K-1]$ is the direction index.

The range of view $\mathcal{R} = \{r_0, \dots, r_{N-1}\}$ is the discrete set of ranges visualized in the frames generated. With the sampling rate f_s and a defined time window for signal reception with the boundaries T_{Start} and T_{Stop} , the number of acquired samples is given by

$$N = \lfloor (T_{\text{Stop}} - T_{\text{Start}}) f_s \rfloor + 1. \quad (6)$$

The range corresponding to the n th sample is calculated with

$$r_n = c \cdot \frac{nT_s + T_{\text{Start}}}{2} \quad (7)$$

where c is the speed of sound, $n \in [0, \dots, N-1]$ is the sample index and $T_s = 1/f_s$ is the sampling period.

The transformation of the spherical coordinates range r , azimuth θ and elevation φ to Cartesian coordinates is given by

$$[x, y, z] = [r \sin(\theta) \cos(\varphi), r \sin(\theta) \sin(\varphi), r \cos(\theta) \cos(\varphi)]. \quad (8)$$

Here, $[x, y, z]$ are the right-handed Cartesian coordinates with the xy -plane being the aperture plane, the xz -plane being the horizontal plane and the z -axis pointing in the direction $(\theta, \varphi) = (0^\circ, 0^\circ)$ (Fig. 1). The region of interest is defined once at the very beginning of the frame generation and remains constant for the following three processing stages.

A. Pulse-Echo Sequence

In the pulse-echo sequence, an ultrasonic pulse is transmitted in order to create and capture echo-signals from the scatterers in front of the array.

For transmission, a single transducer is driven with a unipolar 40-kHz square wave pulse with a length of 40 periods ($T_{\text{pulse}} = 1$ ms) and an amplitude of $20 V_{\text{pp}}$. Due to reduced effective transmit aperture size ($\varnothing 3.4$ mm $= 0.4 \lambda$) of the waveguide port compared to the aperture of the bare transducer ($\varnothing 9.4$ mm $= 1.1 \lambda$), a close-to hemispherical wave $\tilde{s}(t)$ is transmitted. Therefore, scatterers in all spatial directions in the region of interest are irradiated with a single pulse. The transmitted complex equivalent pressure signal $\tilde{s}(t)$ is expressed by

$$\tilde{s}(t) = A(t) \cdot e^{j2\pi f_0 t} \quad (9)$$

where $A(t)$ is the amplitude envelope of the pulse, j is the imaginary unit, and $f_0 = 40$ kHz is the signal frequency.

After pulse transmission, all $M = 64$ transducers are used to receive the echo signals. For a single echo signal originating from a point scatterer in the far-field from direction (θ_o, φ_o) , the analog received signal of the m th transducer is given by

$$\tilde{s}_{\text{RX}, m}(t) = \tilde{s}[t - \tau - \Delta t_m(\theta_o, \varphi_o)] \quad (10)$$

where τ denotes the sound propagation time delay depending on the range of the echo origin $r = c \cdot \tau/2$ and $\Delta t_m(\theta, \varphi)$ is the specific time delay of the m th transducer depending on the direction of arrival (θ, φ) , that is,

$$\Delta t_m(\theta, \varphi) = -\left[\frac{x_m}{c} \sin(\theta) \cos(\varphi) + \frac{y_m}{c} \sin(\varphi)\right]. \quad (11)$$

Here, (x_m, y_m) are the m th transducer position coordinates and c is the speed of sound.

Since the relative bandwidth of the signal is small, the narrowband assumption holds [48]. This implies that the m th transducer-specific delay $\Delta t_m(\theta, \varphi)$ is negligible for the amplitude envelope, i.e., $A[t - \tau - \Delta t_m(\theta_o, \varphi_o)] \approx A(t - \tau)$, such that

$$\begin{aligned} \tilde{s}_{\text{RX}, m}(t) &\approx \tilde{s}(t - \tau) \cdot e^{-j2\pi f_0 \Delta t_m(\theta_o, \varphi_o)} \\ &= \tilde{s}(t - \tau) \cdot a_m(\theta_o, \varphi_o) \end{aligned} \quad (12)$$

is valid. In (12), $a_m(\theta, \varphi) = e^{-j2\pi f_0 \Delta t_m(\theta, \varphi)}$ denotes the steering factor of the m th transducer.

The sample acquisition of the received signals starts after the pulse transmission ($T_{\text{start}} = T_{\text{pulse}}$) using a sampling rate of $f_s = 200$ kHz, obtaining N samples per channel (6). The digitized received signal of the m th transducer is given by

$$s_{\text{RX}, m}(n) = \text{Re}\{\tilde{s}_{\text{RX}, m}(nT_s + T_{\text{pulse}})\} \quad (13)$$

where $T_s = 1/f_s$ is the sampling period and n is the sample index. The digitized signals $s_{\text{RX}, m}(n)$ of all M transducers are transferred to the PC for signal processing.

B. Array Signal Processing

The signal processing chain consists of analytic signal conversion, matched filtering, conventional beamforming and envelope extraction. The processing is mainly performed in the frequency domain and is parallelized using a GPU and the Nvidia CUDA API. Frequency domain processing offers three main advantages. First, it reduces the computational load of filtering and analytical signal conversion since no convolution is required. Second, the matrix multiplication for conventional beamforming can be reduced to the narrow frequency band of the received echo spectrum. Last, compared to time-domain delay-and-sum-beamforming, phase-shifting is not constrained to multiples of the sampling period. Although conventional beamforming provides poor resolution compared to other techniques, it enables large volume scanning at high frame rates, as it requires only a single matrix multiplication. In the following, the implementation of the processing chain is described.

After the data have been received by the PC, all signals $s_{\text{RX}, m}(n)$ are buffered in the GPU memory and transformed into the frequency domain using the CUDA batched fast Fourier transform *cuFFT*, that is,

$$\hat{s}_{\text{RX}, m}(l) = \mathcal{F}_N\{s_{\text{RX}, m}(n)\} \quad (14)$$

where $\mathcal{F}_N\{\cdot\}$ denotes the vectorwise N -point Fourier transform from time to frequency domain and $l \in \{0, \dots, N-1\}$ is the frequency bin index.

Next, each frequency domain signal of $\hat{s}_{\text{RX}}(l)$ is converted to an analytic signal and multiplied with the frequency response of the matched filter. The resulting signal for the m th transducer is given by

$$\hat{s}_{\text{BP}, m}(l) = \hat{u}(l) \cdot \hat{h}(l) \cdot \hat{s}_{\text{RX}, m}(l). \quad (15)$$

Here, $\hat{u}(l)$ represents the conversion function for creating the analytic signal in the frequency domain, that is,

$$\hat{u}(l) = \begin{cases} 1, & \text{if } l = 0 \\ 2, & \text{if } 0 < l \leq \frac{N}{2} \\ 0, & \text{otherwise} \end{cases} \quad (16)$$

and $\hat{h}(l)$ is the frequency response of the matched filter.

In order to determine the originating directions of the impinging echoes, the preprocessed signals $\hat{s}_{\text{BP}, m}(l)$ are spatially filtered using a conventional beamformer [48], evaluating K different scanning directions defined by the field of view $\mathcal{P}$ (2). This multievaluation is implemented as parallel tiled matrix multiplication, that is,

$$\hat{\mathbf{S}}_{\text{BF}} = \mathbf{W}^H \cdot \hat{\mathbf{S}}_{\text{BP}} \quad (17)$$

where $\mathbf{W}^H$ is the $[K \times M]$ beamformer matrix, $\hat{\mathbf{S}}_{\text{BP}}$ is the $[M \times N]$ preprocessed input data matrix and $\hat{\mathbf{S}}_{\text{BF}}$ is the resulting $[K \times N]$ spatially filtered output data matrix. The single elements of the respective matrices are given by

$$\mathbf{W}_{(k, m)}^H = w_m^*(\theta_k, \varphi_k) \quad (18)$$

$$\hat{\mathbf{S}}_{\text{BP}, (m, l)} = \hat{s}_{\text{BP}, m}(l) \quad (19)$$

$$\hat{\mathbf{S}}_{\text{BF}, (k, l)} = \hat{s}_{\text{BF}}(\theta_k, \varphi_k, l) \quad (20)$$

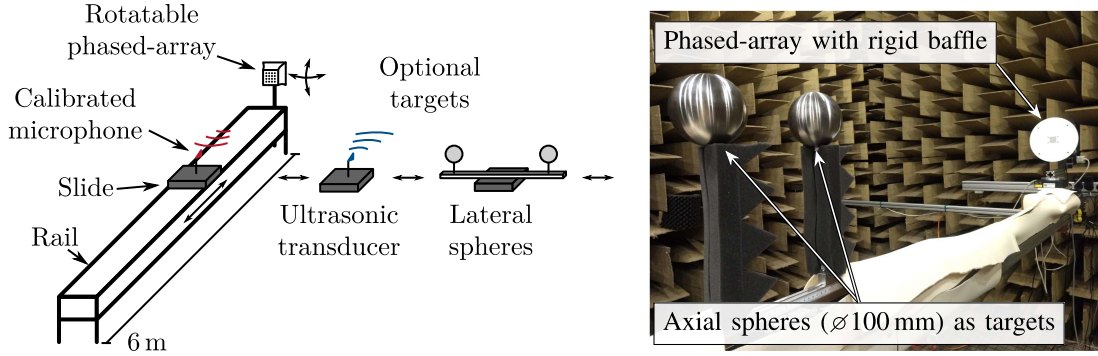


Fig. 3. Measurement setups in the anechoic chamber. The phased-array can be rotated mechanically with two DOF using the rotational axes. For the transmit characteristic measurement, a calibrated microphone is used as the target. For measuring the receive characteristic, the target is replaced with an ultrasonic transducer. Otherwise one or two spheres with a diameter of 100 mm are used, which are positioned axially or laterally adjacent.

where $w_m^*(\theta, \varphi)$ is the beamforming factor of the m th transducer, corresponding to the complex conjugate of its steering factor, that is,

$$w_m^*(\theta, \varphi) = a_m^*(\theta, \varphi) = e^{j2\pi f_0 \Delta t_m(\theta, \varphi)}. \quad (21)$$

For further optimization, the matrix multiplication in (17) can be evaluated only for the frequency bins, containing the echo spectrum.

After multiplying the matrices, all K spatially filtered signals in the field of view are obtained and transformed back into the time domain. Finally, the amplitude envelopes $s_{\text{Env}}(\theta_k, \varphi_k, n)$ are determined by calculating the absolute value of each element, since the analytical signals have already been created in (15), that is,

$$s_{\text{Env}}(\theta_k, \varphi_k, n) = |\mathcal{F}_N^{-1}\{\hat{s}_{\text{BF}}(\theta_k, \varphi_k, l)\}| \quad (22)$$

where $\mathcal{F}_N^{-1}\{\cdot\}$ denotes the vectorwise N -point inverse Fourier transform using the CUDA batched IFFT. The amplitude envelopes correspond to the directional and distant dependent echo amplitudes, which are graphically displayed to create an image of the space in front of the array.

C. Visualization

In the visualization stage, the horizontal sectional image (B-Scan) and the 3-D volume image (3-D-Scan) are generated with OpenGL using the processed amplitude envelope data. First, the corresponding position of each amplitude value $s_{\text{Env}}(\theta_k, \varphi_k, n)$ is given by the direction index k to obtain the field of view direction $\mathbf{p}_k = (\theta_k, \varphi_k)$ (5) and the sample index n to obtain the range r_n (7). The determined coordinates r_n, θ_k, φ_k are transformed to Cartesian coordinates using (8). Then, each amplitude value is color-coded and rendered at its specified coordinates. The image generated is linearly interpolated to fill blank voxels. In addition, weak amplitude voxels in the 3-D-Scan are made transparent using alpha blending.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

The HP method is analyzed in an anechoic chamber with emphasis on the transmit and receive characteristics, lateral

and axial resolution, range of view and field of view, as well as the achievable frame rates. Regarding these properties, the HP method is compared to the DTB method. For both methods the same phased-array and measurement setup are used.

A. Measurement Setup

In the anechoic chamber, the phased-array is attached to two rotational axes at the end of a 6-m-long rail. On the rail there is a movable slide on which different targets can be mounted. Depending on the measurement, a calibrated microphone (B&K Type 4138), an ultrasonic transducer (Murata MA40S4S), one or two hollow steel spheres ($\varnothing 100$ mm) are used as targets. The spheres are mounted either axially or laterally next to each other. Using the rotational axes and the slide, the targets can be positioned freely in the coordinate system of the phased-array (Fig. 3).

B. Transmit and Receive Characteristic

The transmit and receive characteristic describe the spatial sound distribution of the transmit pulse and the spatial filter response of the conventional beamformer, respectively. Important parameters are the -3 -dB main lobe width and the sidelobe level.

For the transmit characteristic, a broad, uniform main lobe is desirable. This allows scatterers to be irradiated equally to generate echoes from all spatial directions with a single transmit pulse. The receive characteristic requires a narrow main lobe in order to determine the originating directions of the received echoes. Side lobes cause phantom reflections which degrade the image quality and result in false detections. Therefore, a low sidelobe level is preferred.

For the transmit characteristic measurement the calibrated microphone is used as target and mounted at a range of 2 m on the slide in front of the phased-array. The phased-array sequentially transmits pulses using a single transducer (Fig. 4). The calibrated microphone records the incoming pulse signals and their maximum amplitudes are stored. After each pulse transmission, the phased-array is rotated mechanically in steps of 2° using the rotational axes. This pitch-catch sequence is repeated until all directions of [Fig. 5(a)] have been covered.

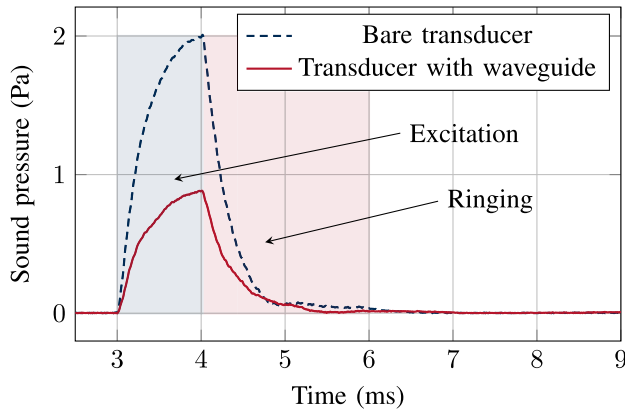


Fig. 4. Measured amplitude envelope of two pulses ($T_{\text{Pulse}} = 1$ ms) transmitted by a bare transducer and by the same transducer inserted into the waveguide, respectively. Both pulses are recorded by a calibrated microphone at a range of 1 m, directly in front of the transducer ($0^\circ, 0^\circ$). Comparing both pulses, the overall amplitude reduction is -7.1 dB consisting of -3.8 -dB waveguide losses due to internal reflections and friction, plus -3.3 -dB diffraction loss, since the directivity pattern is widened (Fig. 5). Apart from attenuation, the shape of the pulse is preserved, as the waveguide has no significant impact on the bandwidth.

The measurement is normalized to the largest received amplitude and is linearly interpolated (Fig. 5).

The measured transmit characteristic has a -3 -dB main lobe width of $\pm 50^\circ$. There are several narrow side lobes due to the pressure-release piston radiator directivity pattern, since the rigid baffle is finite and additionally interrupted by the non-transmitting neighboring waveguide ports. The minimum relative amplitude at -90° is -12 dB or 25% of the largest value measured, respectively. The reduction of the effective transmit aperture by the waveguide leads to a more uniform transmit characteristic compared to the bare transducer [Fig. 5(b)]. However, widening the transmit characteristic also results in a weakened maximum pulse amplitude due to diffraction loss of -3.3 dB, as expected. In addition, the waveguide channel itself attenuates the transmitted pulse by -3.8 dB due to internal reflections and friction (Fig. 4). Apart from the attenuation, the shape of the pulse is preserved, since the waveguide has no significant impact on the bandwidth of the pulse, as the piezoelectric transducer is narrowband.

The receive characteristic is measured by using an ultrasonic transducer as target, which is positioned at a range of 2 m on the slide in front of the phased-array. The transducer is of the same type as the array elements (MA40S4S, 40 kHz, $\varnothing 9.6$ mm) and can be considered to be a point source at this range. The phased-array is in receive-only mode and is not rotated mechanically during this measurement. Instead, it points directly to the transducer. The transducer transmits a single pulse, which is recorded by all phased-array transducers. Using this recorded data set, the conventional beamformer generates a summed, spatially filtered signal for each direction shown in [Fig. 6(a)]. The receive characteristic shows the normalized maximum amplitudes of these signals as a function of the direction (Fig. 6).

The measured receive characteristic has a main lobe width of 12° and a sidelobe level of -11.5 dB. Compared to the

analytic model (12° , -12.8 dB), the main lobe width is in good agreement. However, the measured receive characteristic has a sidelobe level 1.3 dB higher and its shape is asymmetrical. This is caused by differences in the sensitivities and phase responses between the ultrasonic transducers, also affecting the lateral resolution, as shown in Section IV-C.

C. Axial and Lateral Resolution

The axial and lateral resolutions describe the minimum distance between two objects such that they appear separated in the image generated. The objects are defined as separable if their echoes create two distinct local maxima.

In order to measure the axial or lateral resolution, two steel spheres ($\varnothing 100$ mm) are placed either axially or laterally next to each other on the profile mounted on the slide. The slide is positioned at a range of $r = 2$ m in front of the phased-array. The distance D between the spheres is gradually increased. After each increment, the separability is evaluated using B-Scan images (Fig. 7). The minimum distance $D_{\text{min, ax}}$ for separability of two axially adjacent spheres directly determines the axial resolution $D_{\text{res, ax}}$, that is,

$$D_{\text{res, ax}} = D_{\text{min, ax}} \cdot \quad (23)$$

However, the lateral resolution $\theta_{\text{res, lat}}$ is expressed as an angle, since the minimum distance $D_{\text{min, lat}}$ for separability of two laterally adjacent objects depends on the range r to the phased-array, that is,

$$\theta_{\text{res, lat}} = 2 \cdot \arctan\left(\frac{D_{\text{min, lat}}}{2 \cdot r}\right). \quad (24)$$

The axial resolution is affected by two factors, the pulse-length and the created sound shadow depending on the object shape. For the measurement setup described, the axial resolution is $D_{\text{res, ax}} = 200$ mm for both, the HP and the DTB method, as the excitation time is the same in both methods ($T_{\text{Pulse}} = 1$ ms). Due to the narrow bandwidth of the transducer resulting in an increased pulselength (Fig. 4), the measured results are higher compared to the ideal axial resolution of a 1-ms pulse with minimum rise and fall times (171.5 mm). The measurement further shows that a direct line of sight is not necessarily required for object detection. Since sound waves can diffract around objects, the echo of the rear object still reaches the array, but its amplitude is weakened due to the sound shadow of the front object.

The measured lateral resolution using the DTB method is improved ($\theta_{\text{res, lat}} = 12^\circ$) compared to the HP method ($\theta_{\text{res, lat}} = 14^\circ$). The reason is that beamformed transmit pulses irradiate the region of interest direction-selectively. Therefore, they provide another layer of spatial filtering during pulse transmission, in addition to the conventional beamformer after echo reception. Compared to the achievable lateral resolution using analytic models (DTB 9° , HP 11°), the measured results show a loss of 3° in both cases. Again, the cause is variations in the sensitivities and phase responses between the transducers. In order to correct these variations, an array calibration is required, which will be part of future work.

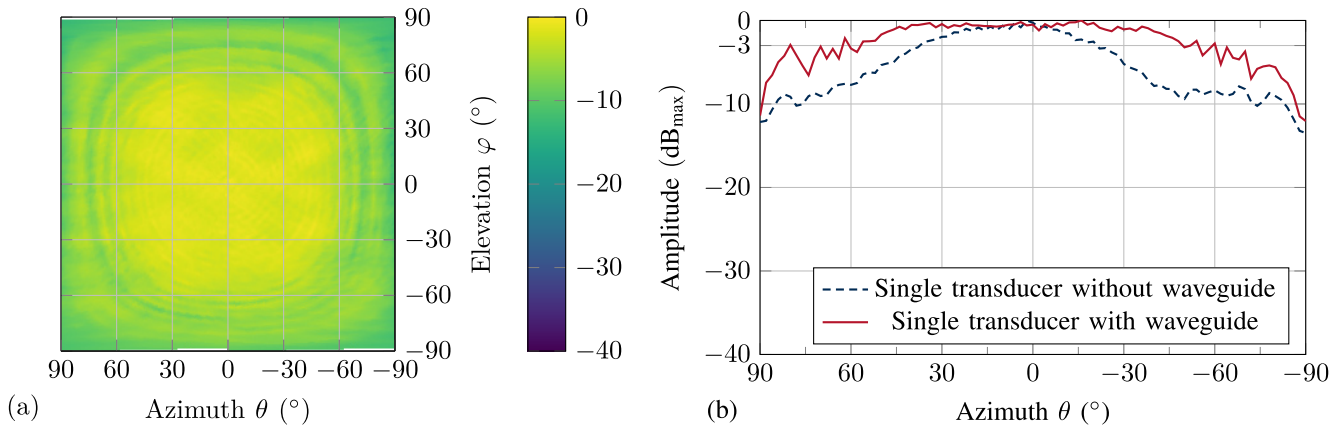


Fig. 5. (a) Measured HP method transmit characteristic and (b) its horizontal sectional view of a single transmitting transducer, which is inserted into the waveguide. In addition, the sectional view in (b) is compared to the transmit characteristic of a single transducer without waveguide. For these measurements, the phased-array is mechanically rotated incrementally with the rotational axes. At each step, a pulse is transmitted, which is recorded by the calibrated microphone at a range of 2 m. The maximum amplitudes normalized to the corresponding largest value of these pulses are shown. There are several sidelobes due to the pressure-release piston radiator directivity pattern, since the rigid baffle is finite and additionally perforated by the nontransmitting waveguide ports. The transmit characteristic of the transducer with the waveguide attached is more uniform since the effective aperture is reduced.

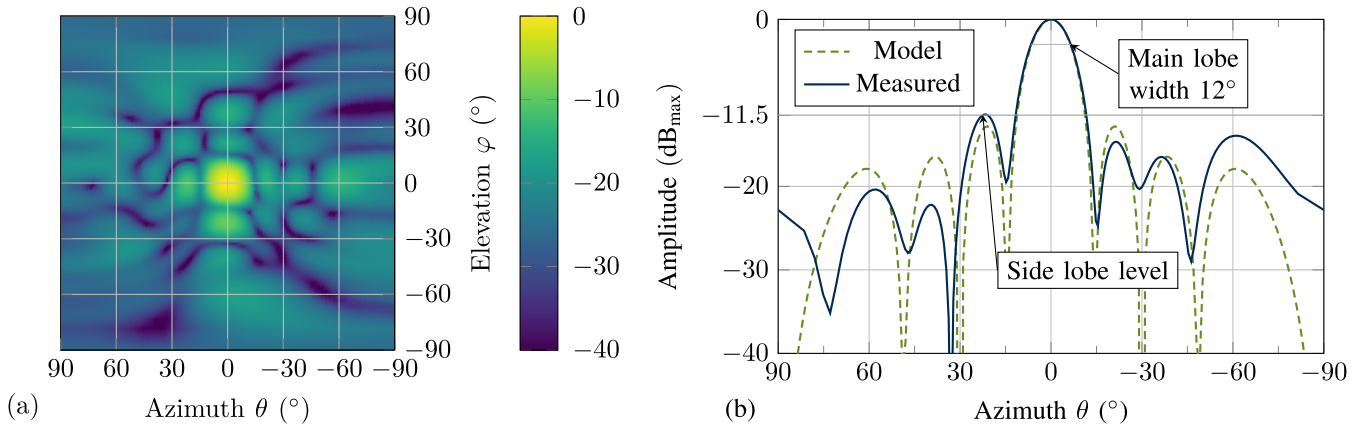


Fig. 6. (a) Measured receive characteristic and (b) its horizontal sectional view. For this measurement, an ultrasonic transducer transmits from a range of 2 m directly to the only receiving phased-array. The conventional beamformer generates the reception pattern shown, which corresponds to the transfer function of the spatial filter for the direction (0°, 0°). The asymmetry and irregularities compared to the analytic model are caused by differences in the sensitivities and phase responses of the individual transducers. Since both methods use all transducers for receive beamforming, the corresponding receive characteristics are equal.

D. Range of View and Field of View

In the B-Scan images shown, there is a blind zone up to a range of 1 m (Fig. 7). In this zone, high amplitudes are detected, which are caused by the transmitted pulse and the ringing of the transducer. These high amplitudes mask incoming echoes. With a suitable amplitude scaling of the colorbar for close targets, echoes above a minimum range of $r_{\min} = 0.5$ m are detectable with both transmission methods. Since the amplitudes in the blind zone change only slightly between several pulse excitations, they can be removed by subtraction with a prerecorded image. For characterization, the blind zone removal is not active.

To determine the maximum detection range $r_{\max}$, the range r between the phased-array and a single sphere ($\varnothing 100$ mm) is gradually increased along the rail. After each increment, a B-Scan image is used to evaluate whether the echo of the sphere is detectable. The echo is defined as detectable if its amplitude is higher than the noise level. The maximum

detection range using the DTB method is larger ($r_{\max} > 6$ m) than using the HP method ($r_{\max} = 3$ m). Since the DTB method uses all transducers for pulse transmission, the amplitude of the beamformed pulse is approximately 30 times higher than the amplitude of the single HP. Therefore, the beamformed pulse and its echos can propagate further before being attenuated by air below the noise level. For the DTB method, the validation of the maximum range of 6 m is limited by the anechoic chamber size.

The same procedure is used to determine the maximum field of view. However, not the range of the sphere is varied ($r = 2$ m), but its direction in the array coordinate system. For this purpose, the phased-array is rotated mechanically in 5° steps with the horizontal rotational axis. For both methods the maximum echo amplitude decreases with increasing direction due to the directivity of a single transducer. Using the DTB method, the echo of the sphere is detectable in the entire field of view of $\pm 90^\circ$. With the HP method, the field of view is limited to $\pm 80^\circ$ due to the overall weaker echo amplitude.

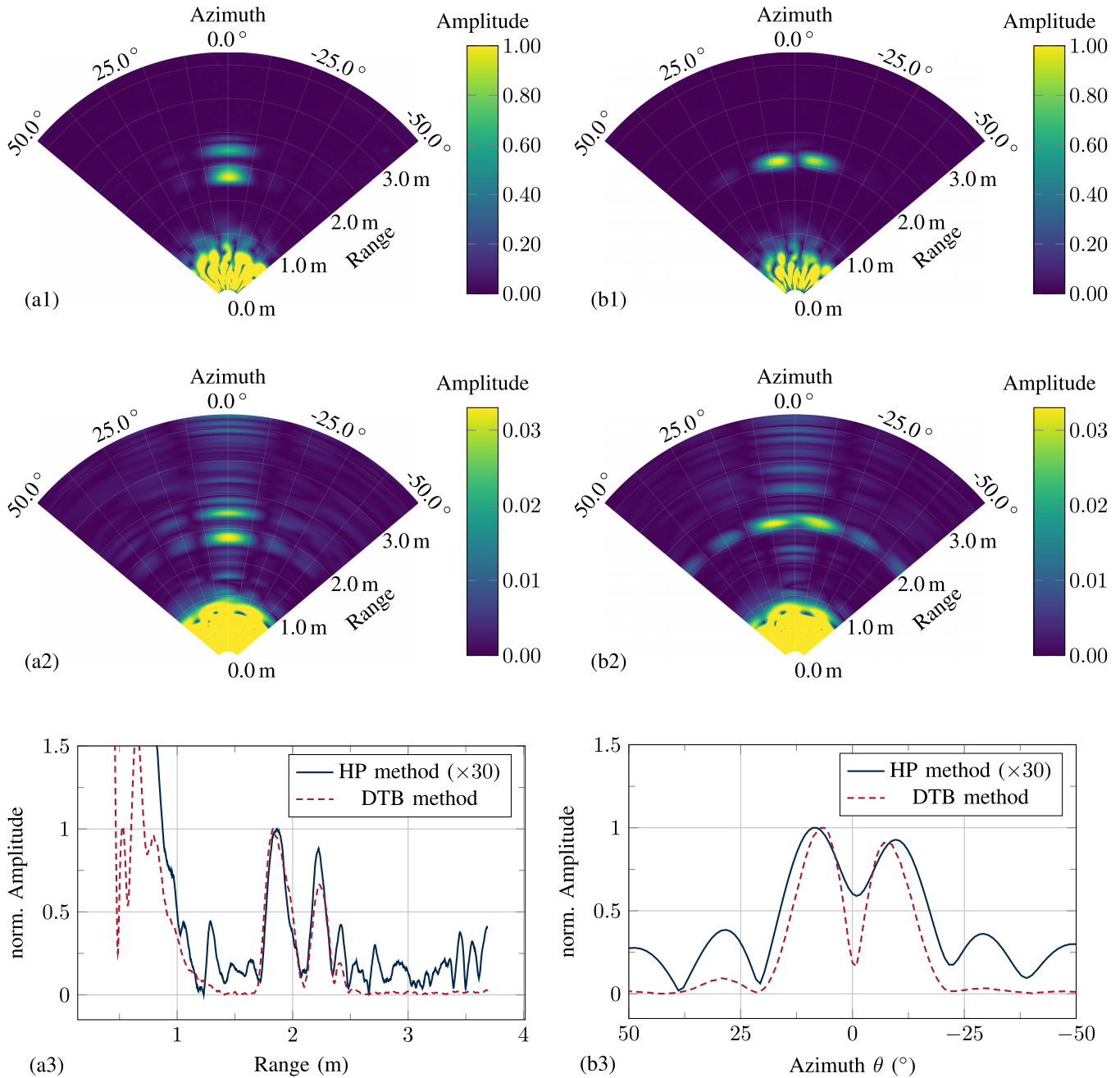


Fig. 7. B-Scans of the axial resolution measurement using (a1) DTB method and (a2) HP method and (a3) their corresponding axial sections at 0° . The B-Scans show the echos of two spheres ($\varnothing 100$ mm) positioned axially one behind the other at a range of 1.8–2.2 m, respectively. Although there is no direct line of sight, the echos can be localized with both methods. Since the amplitude of the transmitted pulse using the HP method is 30 times weaker, the SNR is decreased compared to the DTB method. The high amplitudes of the transmitted pulse and the ringing transducers mask incoming echos in a range of up to 0.5 m, also referred to as blind zone. In this zone, objects are not detectable without further processing. B-Scans of the lateral resolution measurement using (b1) DTB method and (b2) HP method and (b3) their corresponding lateral sections at 2.1 m. The B-Scans show the echos of two spheres ($\varnothing 100$ mm) positioned laterally adjacent in the directions 7° and -7° , respectively. Both methods generate a distinct local maxima for each sphere, such that they are separable. For the HP method, the local minimum between the spheres is higher and the side lobes are more prominent due to the wider main lobe and higher sidelobe level of its point spread function.

E. Frame Rates

The frame rate indicates the number of completely generated and rendered B-Scans or 3-D-Scans within one second. The unit of the frame rate is frames per second (FPS). It is equivalent to the pulse repetition frequency (PRF) when using the HP method. However, it differs from the PRF when using the DTB method since multiple pulses are required to generate an image.

For comparison of the 3-D-Scan frame rates of both methods, the region of interest is defined with

$$\begin{aligned} \theta_{\min} &= \varphi_{\min} = -50^\circ, & \theta_{\text{step}} &= \varphi_{\text{step}} = 2^\circ \\ \theta_{\max} &= \varphi_{\max} = 50^\circ, & \mathcal{R} &= \{0.017 \text{ m}, \dots, 3 \text{ m}\} \end{aligned} \quad (25)$$

where $\theta_{\min}$, $\theta_{\max}$, $\varphi_{\min}$, and $\varphi_{\max}$ are the minimum and maximum azimuth and elevation angles, θ_{step} and φ_{step} are the angular step sizes and $\mathcal{R}$ is the range of view.

TABLE I
SUMMARY OF THE MEASUREMENT RESULTS FOR THE HP AND DTB METHOD

	HP method	DTB method	Remark
TX main lobe width	100°	21°	
RX main lobe width	12°	12°	
TX side lobe level	-	-7.2 dB	
RX side lobe level	-11.5 dB	-11.5 dB	
Axial resolution	200 mm	200 mm	Targets: two steel spheres ($\varnothing$ 100 mm), pulse length 1 ms
Lateral resolution	14°	12°	Targets: two steel spheres ($\varnothing$ 100 mm)
Min. detection range	0.5 m	0.5 m	Due to blind zone
Max. detection range	3 m	> 6 m*	Target: one steel sphere ($\varnothing$ 100 mm), *limited by anechoic chamber size
Field of view	$\pm 80^\circ$	$\pm 90^\circ$	Target: one steel sphere ($\varnothing$ 100 mm) at 2 m
B-Scan frame rate	43 FPS	0.76 FPS	Number of directions: 51, max. range of view: 3 m
3D-Scan frame rate	29 FPS	0.015 FPS	Number of directions: 2601, max. range of view: 3 m

For the B-Scan frame rate comparison, the region of interest parameters are identical, however, the field of view is limited to the horizontal plane, i.e., $\varphi_{\min} = \varphi_{\max} = 0^\circ$. Thus, the numbers of directions to be evaluated for generating a 3-D-Scan K_V and a B-Scan K_B are

$$K_V = 51 \cdot 51 = 2601 \text{ and } K_B = 51. \quad (26)$$

The frame rate $f_{\text{Frame}} = 1/T_{\text{Frame}}$ is calculated by measuring the time T_{Frame} between the first pulse transmission and the complete generation and rendering of the image. The frame rates given in (Table I) were averaged over 20 frames. Using the HP method, B-Scan and 3-D-Scan frame rates of 43 and 29 FPS are achievable. The frame rates are mainly determined by the sound propagation time (B-Scan 75%, 3-D-Scan 50%). The remaining part consists of data transfer and processing time. With the DTB method, the frame rates for a B-Scan and 3-D-Scan are 0.76 and 0.015 FPS, respectively. Since this method sends a pulse for each direction, the achievable frame rates significantly decrease with increasing number of directions. However, this relation also indicates that the DTB method can achieve higher frame rates if the field of view is reduced or the angular step size is increased. The system is able to select the region of interest and the transmission method dynamically.

V. CONCLUSION

In this article, we introduced and evaluated an air-coupled ultrasonic 3-D imaging system using a URA based on a waveguide and efficient low-cost transducers. In contrast to similar air-coupled ultrasonic imaging systems (Section I), this approach allows the adaptation of multiple transmit and receive methods with individual strengths and weaknesses. For instance, the experimental characterization demonstrated that the system is capable of localizing multiple objects in real-time with the HP method, although there are trade-offs compared to the DTB method, most notably the reduced range of view. Since the system supports both methods, the appropriate variant can be selected dynamically depending on the situation. Considering that a sufficient frame rate is required, the HP method is more suitable for close range situations where a wide field of view is mandatory, whereas the DTB method is an appropriate choice for generating images with a long range of view but only a limited field of view. Given this adaptability, ultrasonic 3-D imaging has high potential for improving the

robustness of in-air environmental monitoring in conjunction with lidar, radar and camera systems. Future work is aimed at the enhancement of the angular resolution and image quality by calibrating transducer variations, evaluating nonuniform array topologies and utilizing image processing algorithms.

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